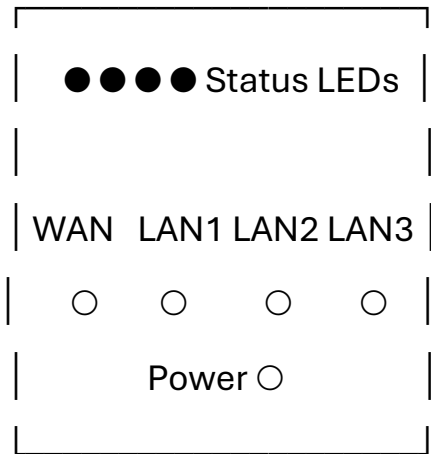
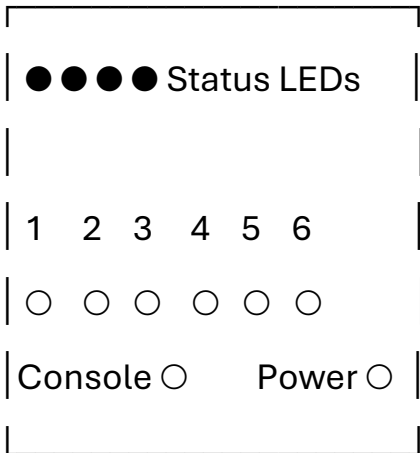


Task 1: Router and Network Switch Diagram

ROUTER



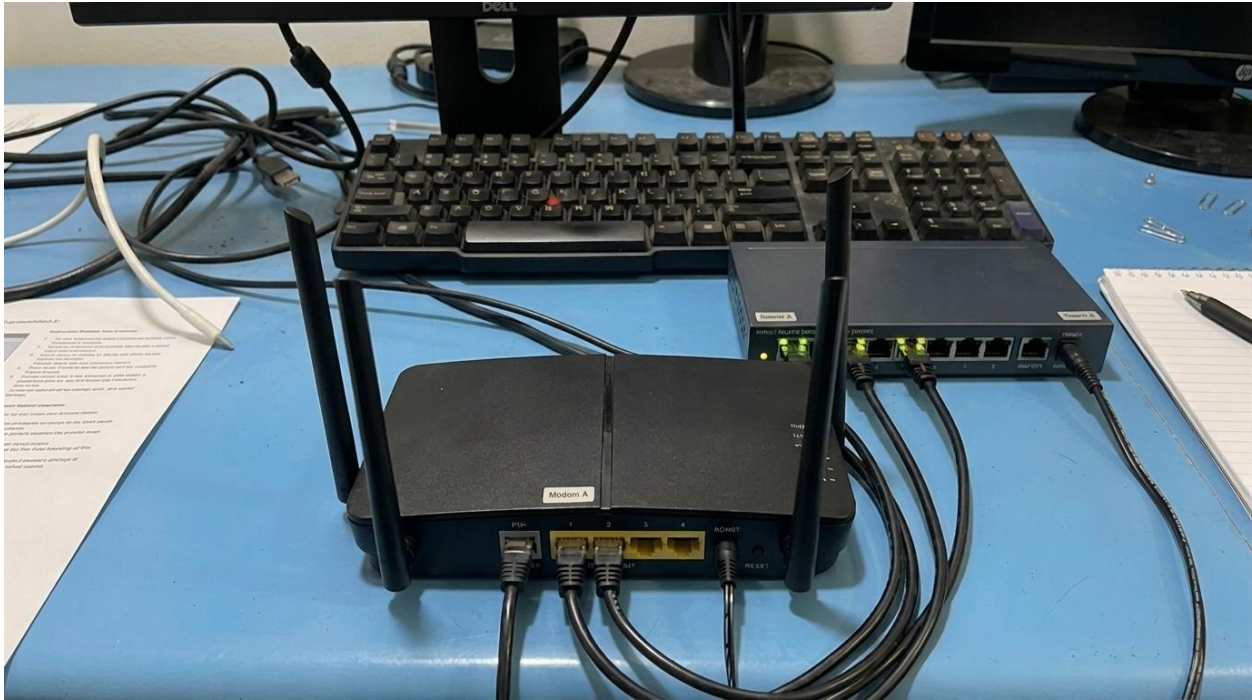
SWITCH



Task 2: Key Differences Between Router and Switch

Difference	Router	Switch
Main Function	Connects different networks and routes traffic between them.	Connects devices within the same local network (LAN).
Address Used	Uses IP addresses to decide where to send data.	Uses MAC addresses to forward data to the correct device.
Networks Connected	Connects different networks, such as a LAN to the Internet.	Mainly connects devices within one LAN.

Task 3: Identify Router and Switch Ports & LEDs



- Power Port: Provides power to the router.
- WAN Port: Connects the router to the modem/ISP.
- LAN Port 1: Connects a wired device to the router.
- LAN Port 2: Connects another wired device to the router.
- LED Indicators: Show power, Internet connection, and network activity.



Switch:

- Power Port: Provides power to the switch.
- Ethernet Port 1: Connects a network device to the switch.
- Ethernet Port 2: Connects another network device to the switch.
- LED Indicators: Show power, link, and network activity.

Task 4: Small Gaming Cafe Network Scenario

I would connect the router first to the ISP line because the router connects the café network to the Internet. Then, I would connect the router to an 8-port network switch using an Ethernet cable. The 8 gaming PCs would then be connected to the switch using separate Ethernet cables. The router would manage Internet traffic using IP addresses, while the switch would forward data between the gaming PCs using MAC addresses. This setup would provide a stable wired network for IPL streaming and multiplayer gaming.